

Method

1. Study method and design

This study employs a multicenter cross-sectional design, which has lots of advantages. Cross-sectional studies are time-saving by eliminating longitudinal follow-up. A snapshot of disease prevalence or risk factors is provided to guide preventive and therapeutic strategies. Additionally, it can analyze multiple exposures and disease and test or suggest hypotheses about determinants and correlates of health conditions. Furthermore, it can do comparative analysis across diverse populations.

However, cross-sectional designs present some limitations. Primary among these is the inability to establish temporal relationships between exposure and outcome. And it is difficult to estimate past exposure, especially if it occurred long time ago. Additionally, reporting or recall bias is sensitive if exposures are subjective, and response rates and representativeness of prevalence of a disease in population is sensitive.

2. Sampling, data collection , exposure and bias

The target population of this study comprised COPD patients consecutively enrolled in outpatient settings across Spain by 225 internal medicine specialists (enrolled 866 patients finally). Eligible participants met three criteria: spirometry-confirmed COPD diagnosis ≥ 6 months prior; cumulative smoking history ≥ 10 pack-years; no history of COPD exacerbation requiring corticosteroid or antibiotic treatment within 4 weeks before enrollment.

The study utilized a stratified sampling method, with the sampling frame comprising outpatient medical records from 225 internists nationwide in Spain.

COPD is characterized by an abnormal inflammatory response of the lungs to noxious particles or gases. In this study, COPD severity was classified into four stages based on forced vital capacity in 1 second (FEV_1)(Figure 1).

FEV ₁	COPD severity
FEV ₁ ≥80%	Stage I mild
50% ≤ FEV ₁ <80%	Stage II moderate
30% ≤ FEV ₁ <50%	Stage III severe
FEV ₁ <30%FEV ₁ or <50% (chronic respiratory failure)	Stage IV very severe

Figure 1: COPD severity

This study examined two exposures: comorbidity and smoking. Comorbidities were assessed through two distinct methodologies: a predefined list of 24 disease, and the Charlson Index, calculated as both unadjusted and age-adjusted scores (Figure 2). Smoking history required a minimum of 10 pack-years, based on self-reported histories and medical record verification.

Charlson Index	comorbidity
0-1	no comorbidity
2	low comorbidity
≥3	high comorbidity

Figure 2: Comorbidity

This study is subject to three primary biases. Firstly information bias, charlson Index miss important comorbidities and comorbidity data came from medical records and self-reports. Secondly, selection bias, study participants came solely from internal medicine clinics. Thirdly, recall bias, patient-reported smoking history and past health conditions may be inaccurate.

3 Confounding analysis and alternative method

The confounders of this study are age, pack-years of smoking, hemoglobin level, and dyspnea. In univariate analysis, the unadjusted and age-adjusted Charlson scores

were analyzed in the same time. The scores of unadjusted and age-adjusted differed significantly, indicating age as a confounder. The multivariate linear regression (Figure 3) analyzed associations between Charlson score (dependent variable) and COPD severity, age, smoking pack-years, hemoglobin levels, and dyspnea to evaluate their independent influence on the Charlson score. Results showed no significant association between COPD severity and Charlson score, while age, pack-years, hemoglobin, and dyspnea remained significantly associated.

Independent variable	P- value	Significant associated
COPD severity	0.13	No
Age	<0.0001	Yes
Smoking pack-years	0.03	Yes
Hemoglobin levels	<0.0001	Yes
Dyspnea	0.04	Yes

Figure 3: Multivariate linear regression

To address the limitations of the current cross-sectional study design, there are two alternative methodologies: prospective cohort studies and nested case-control designs. Prospective cohort studies enable longitudinal tracking to clarify temporal relationships and reduce reverse causation. Nested case-control designs within cohorts allow efficient analysis of incident comorbidities via matched controls.

World count: 500 (excluding figures).

Reference:

Echave-Sustaeta, J. M., Casanova, L. C., Cosio, B. G., Soler-Cataluña, J. J., Garcia-Lujan, R., & Ribera, X. (2014). Comorbidity in chronic obstructive pulmonary disease. Related to disease severity? *International Journal of COPD*, 9, 1307–1314. <https://doi.org/10.2147/COPD.S71849>